

Atahar Hossain Piash

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Education

B.Sc. in Computer Science and Engineering

2022 – Present

North South University, Dhaka, Bangladesh

- **Major:** Artificial Intelligence & Software Engineering
- **CGPA:** 3.74 / 4.00
- **Relevant Coursework:** Machine Learning, Data Structures & Algorithms, Database Systems, Software Engineering, Operating Systems, Web Programming, Embedded Systems

Higher Secondary Certificate (HSC), Science

2021

Dhaka Imperial College, Dhaka

- **GPA:** 5.00 / 5.00

Secondary School Certificate (SSC), Science

2018

Armanitola Govt. High School, Dhaka

- **GPA:** 4.83 / 5.00

Research Experience

NSU Trimodal AI Lab: Under Graduate Research Assistant

2025 – Present

Advisor: Dr. Md. Adnan Arefeen, Assistant Professor, Department of ECE, NSU

- **Bengali Mental Health Conversational Agent**
- Developed a privacy-first Bengali conversational AI system focused on empathetic, context-aware responses and low-compute local deployment; fine-tuned compact LLMs using parameter-efficient LoRA and evaluated model performance using BLEU, BERTScore, and human evaluation.
- **Technologies:** Python, PyTorch, Hugging Face Transformers, LoRA/PEFT, NLP, LLMs, Django, REST APIs, SQLite
- **AdaVideoCap : Adaptive Multi-Model Video Captioning**
- Contributed to a multimodal video understanding framework that dynamically routes video frames across multiple VLMs; designed and evaluated temporal grouping and Temporal-GNN strategies involving frame sampling, temporal mosaicing, parallel processing, and entity-level tracking for efficient long-video and driver-monitoring analysis.
- **Technologies:** Python, Computer Vision, VLMs, Temporal Modeling, GNNs, Parallel Processing, Optimization

Selected Projects

Predicting Future E-Commerce Purchasing Behavior

Machine Learning Project

- Developed a machine learning pipeline to predict customer's future purchasing behavior from historical e-commerce activity. Performed data preprocessing, exploratory data analysis, feature engineering, model training, and comparative evaluation.
- **Technologies:** Python, Pandas, NumPy, Scikit-learn, Matplotlib

AI Spam & Phishing Email Detector

NLP & Machine Learning

- Built an NLP-based email classification system using **82,000+ real-world emails across six categories** to identify spam and phishing-related messages.

- Developed an interactive analysis workflow providing classification verdict, confidence score, risk level, and interpretable warning signs.
- Integrated the trained classifier into a polished web interface with an interactive glassmorphism-based user experience.
- **Technologies:** Python, NLP, Machine Learning, Scikit-learn, HTML, CSS, JavaScript

NutriAI: AI Nutrition Assistant

AI-Powered Web Application

- Developed an AI-assisted nutrition platform that generates personalized meal plans and provides nutrition-related guidance.
- Implemented BMI tracking, meal planning, OCR-based input processing, conversation history, and persistent user records.
- **Technologies:** Python, Django, PyTorch, Qwen LLM, Pytesseract, SQLite, REST APIs, HTML/CSS

Genetic Algorithm Game AI

Game AI & Optimization

- Developed a game-playing AI using genetic algorithms to evolve agent behavior through population-based optimization.
- Implemented fitness evaluation, selection, crossover, mutation, and iterative behavioral improvement.
- **Technologies:** Python, Genetic Algorithms, Game AI, Optimization

AI Pac-Man

Game AI

- Developed a Python-based Pac-Man game incorporating autonomous AI decision-making, pathfinding, and environment interaction.
- Implemented game logic and agent behavior without relying on external LLM or API services.
- **Technologies:** Python, Game Development, AI, Pathfinding

Technical Skills

Programming Languages : Python, Java, C, PHP, SQL, Assembly, HTML/CSS

AI & Machine Learning : Machine Learning, Deep Learning, NLP, Computer Vision, LLMs, VLMs, Transformers, Fine-Tuning, LoRA/PEFT, Generative AI

AI Research : Multimodal AI, Temporal Modeling, GNNs, Agentic Systems, LLM/VLM Orchestration, Explainable AI, Resource-Constrained AI

Software Engineering : OOP, SOLID Principles, Design Patterns, Data Structures & Algorithms, Unit & Integration Testing, Debugging, Refactoring, REST APIs, UML, System Design

Backend & Databases : Django, PHP, MySQL, SQLite, Database Design

Tools & Platforms : Git, GitHub, VS Code, PyTorch, Hugging Face Transformers, Ollama, LaTeX, Canva, Adobe Illustrator

Research Interests

Quantum Machine Learning, Agentic Systems, Explainable AI (XAI), Computer Vision, Multimodal AI, Resource-Constrained AI, AI Systems Architecture

Extracurricular Activities

IEEE NSU Student Branch

2023 – Present

General Member; Graphics Team Member

- Participated in technical workshops, seminars, and community outreach programs.
- Contributed to event organization, visual design, and promotional materials for IEEE NSU activities.

NASA Space Apps Challenge — Climate Odyssey

2024

UI/UX Designer & Team Member

- Contributed to the design and prototyping of a collaborative 3D game developed using Unity.
- Worked on UI/UX design, gameplay ideas, concept validation, and prototype development.

NSU Art & Photography Club

2023 – 2024

General Member; Publication Team Member

- Contributed to publication, creative design, and club activities.
- Participated in workshops, seminars, and community initiatives.

References

Dr. Md. Adnan Arefeen

Assistant Professor, Department of ECE, North South University

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